

CYBERSECURITY

ENTERPRISE PROJECT

Design, Implementation, Testing & Hardening

of an Enterprise Operating System Security Environment

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AGENDA

10-Minute Presentation Overview

01 **Project Introduction**

Goals, team roles & environment setup

02 **Windows Administration**

Active Directory, GPO, Password Policy, Auditing

03 **Linux Administration**

Users, SSH hardening, PAM, logging

04 **Security Architecture**

Network design, zones, attack surface

05 **Attack Simulation**

Nmap scanning, Hydra brute-force, analysis

06 **Hardening & Results**

What worked, what didn't, key takeaways

01 PROJECT INTRODUCTION

OBJECTIVE

Design, implement, test and harden a secure enterprise environment using Windows Server, Ubuntu Linux, and Kali Linux — simulating real-world attack and defense scenarios.



Windows Admin

AD, GPO, Password & Lockout Policy, Audit Logs



Linux Admin

Users, SSH hardening, PAM, Auth Logging



Security Analyst

Nmap scanning, Hydra brute-force attacks



Security Architect

Network design, security zones, hardening

Environment: VirtualBox — Host-Only Network — 3 VMs

02 WINDOWS ADMINISTRATION

Password Policy

- History: 24 passwords
- Max age: 30 days | Min: 1 day
- Min length: 8 chars | Complexity: ON
- Reversible encryption: OFF

Audit Policy (GPO)

- Logon events: Success/Failure
- Account management
- Object access tracking
- Policy change monitoring

Account Lockout

- 3 failed attempts → locked
- Lockout duration: 5 minutes
- Reset counter: 5 minutes
- → Protects against brute force

AD Structure & Events

- OUs: IT_Group, Users, Computers
- Event 4624: Successful logon
- Event 4625: Failed logon
- Event 4768/4769: Kerberos tickets

03 LINUX ADMINISTRATION

1. **User & Group Management**
   `adduser / groupadd / usermod -aG`
   Created user1, admin_group; role-based access control

2. **Secure SSH Config**
   `PermitRootLogin no | PasswordAuthentication yes`
   Edited /etc/ssh/sshd_config — disabled root remote login

3. **PAM Authentication**
   `/etc/pam.d/common-auth`
   Enforced complexity rules — protects against weak credentials

4. **Password Aging**
   `chage -M 30 -m 1 -W 7 user1`
   30-day max age, 1-day min, 7-day warning period

5. **Logging & Monitoring**
   `/var/log/auth.log + rsyslog`
   Tracks sudo usage, SSH sessions, unauthorized access attempts

04 SECURITY ARCHITECTURE

UNTRUSTED ZONE



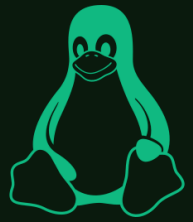
Kali Linux

Attacker Machine

- Nmap scanning
- Hydra brute-force
- Password guessing

ATTACKS

TRUSTED INTERNAL NETWORK (Host-Only)



Ubuntu Server

SSH Port 22: OPEN

- SSH hardened
- Root login disabled
- PAM policies
- Auth logging



Windows Server

Ports 445/3389: FILTERED

- Firewall active
- Domain Controller
- GPO enforced
- Audit enabled

05 ATTACK SIMULATION

1. Service Scanning (Nmap)

Ubuntu → EXPOSED

Target: Ubuntu
Result: Port 22 OPEN
SSH service detected — accessible from Kali

Windows → PROTECTED

Target: Windows
Result: Ports 445, 3389 FILTERED
Firewall blocking all external scans

2. SSH Brute-Force (Hydra)

Ubuntu → VULNERABLE

hydra -l user1 -P test.txt ssh://<ip>
Result: Password FOUND
Weak credentials — user1 was vulnerable

Windows → BLOCKED

hydra over SMB on Windows
Result: FAILED
Missing password file + firewall blocked

3. Unauthorized Access Tests

Ubuntu → PATCHED

SSH login as root
Result: ACCESS DENIED
PermitRootLogin no — config working

Windows → SECURE

Windows SMB / RDP
Result: ACCESS DENIED
Firewall + AD policies effective

06 HARDENING & RESULTS

Linux Hardening

- Root SSH login disabled → prevents direct admin access
- Password complexity via PAM → reduced brute-force success
- Auth logging active → all activity traceable
- Brute-force succeeded with weak password → needs stronger policy
- Password aging configured (30 day max)

Windows Hardening

- Firewall filtered ports 445 & 3389 → all external attacks blocked
- GPO enforced domain-wide password & lockout policy
- Audit logs: 3655+ events captured (4624, 4625, 4768…)
- AD groups limit privilege escalation effectively
- Account locked after 3 failed attempts → brute-force mitigated



KEY TAKEAWAYS

Attack Surface Reduced

Security Layers Applied

CONCLUSION

- Proper system configuration is the first line of defense against attacks.
- Firewalls and filtered ports (Windows) effectively blocked all external intrusion attempts.
- Open ports with weak credentials (Ubuntu SSH) remain a critical vulnerability.
- Security logging is essential for detecting and analyzing threats.
- Defense in depth — combining firewall, auth policies, and monitoring — creates a resilient system.



THANK YOU

Enterprise OS Security — Design, Implementation, Testing & Hardening

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Questions & Discussion